Sequential Quadratic Programming

Consider the equality constrained problem

$$\min_x \; z = f(x) \quad \text{s.t.} \quad c_i(x) = 0 \,, \; i \in \mathcal{E}$$

The Langrangian form

$$\min_{x,p} \; L(x,p) = f(x) - \overbrace{\sum_{i \in \mathcal{E}} p_i c_i(x)}^{p^T c}$$

Consider a Newton Step from (x_k, p_k) to (x_{k+1}, p_{k+1}) which satisfies the stationarity conditions to second order in $L(x,p)$:

- $x_{k+1} = x_k + s_k^x \,, \quad p_{k+1} = p_k + s_k^p \,, \quad s_k = \begin{bmatrix} s_k^x \\ s_k^p \end{bmatrix}$

- $0 = \nabla L(x_{k+1}, p_{k+1}) = \nabla L(x_k, p_k) + \nabla^2 L(x_k, p_k) s_k$

Newton Step approximation

$$\Rightarrow \quad \nabla^2 L(x_k, p_k) s_k = - \nabla L(x_k, p_k)$$

$$\begin{bmatrix} \nabla^2_{xx} L(x_k, p_k) & \nabla^2_{xp} L(x_k, p_k) \\ \nabla^2_{px} L(x_k, p_k) & \nabla^2_{pp} L(x_k, p_k) \end{bmatrix} \begin{bmatrix} s_k^x \\ s_k^p \end{bmatrix} = - \begin{bmatrix} \nabla_x L(x_k, p_k) \\ \nabla_p L(x_k, p_k) \end{bmatrix}$$

$$\begin{bmatrix} \nabla^2_{xx} L(x_k, p_k) & -\nabla c(x_k) \\ -\nabla c(x_k)^T & 0 \end{bmatrix} \begin{bmatrix} s_k^x \\ s_k^p \end{bmatrix} = - \begin{bmatrix} \nabla_x L(x_k, p_k) \\ -c(x) \end{bmatrix}$$

(1) $$\begin{bmatrix} \nabla^2_{xx} L(x_k, p_k) & \nabla c(x_k) \\ \nabla c(x_k)^T & 0 \end{bmatrix} \begin{bmatrix} s_k^x \\ -s_k^p \end{bmatrix} = - \begin{bmatrix} \nabla_x L(x_k, p_k) \\ c(x) \end{bmatrix}$$

$$c(x) \equiv [\, c_1(x) \;\; c_2(x) \;\; \cdots \;\; c_m(x) \,]$$

$$\nabla c(x) \equiv [\, \nabla c_1(x) \;\; \nabla c_2(x) \;\; \cdots \;\; \nabla c_m(x) \,] = J(c)^T$$

The Newton Step expression is also seen as the stationary point solution of the following quadratic program:

(2)
$$\min_{s^x} \quad f(x) + (s^x)^T \nabla_x L(x_k, p_k) + \frac{1}{2}(s^x)^T \nabla_{xx}^2 L(x_k, p_k) s^x$$
$$\text{s.t.} \quad \nabla c(x)^T s^x + c(x) = 0$$

$\Rightarrow$ (1) are the KKT conditions of (2)

Before proceeding further, let's simplify.. Consider the first KKT condition

$$\nabla_{xx}^2 L(x_k, p_k) s_k^x - \nabla c(x_k) s_k^p = -\nabla_x L(x_k, p_k)$$
$$\nabla_{xx}^2 L(x_k, p_k) s_k^x - \nabla c(x_k) s_k^p = -\nabla_x f(x_k) + \nabla c(x_k) p_k$$
$$\nabla_{xx}^2 L(x_k, p_k) s_k^x - \nabla c(x_k)(s_k^p + p_k) = -\nabla_x f(x_k)$$
$$\nabla_{xx}^2 L(x_k, p_k) s_k^x - \nabla c(x_k) p_{k+1} = -\nabla_x f(x_k)$$

$\Rightarrow$ (3)
$$\begin{bmatrix} \nabla_{xx}^2 L(x_k, p_k) & \nabla c(x_k) \\ \nabla c(x_k)^T & 0 \end{bmatrix} \begin{bmatrix} s_k^x \\ -p_{k+1} \end{bmatrix} = \begin{bmatrix} -\nabla f(x_k) \\ c(x_k) \end{bmatrix}$$

Next, noting that when $c(x_k) = 0$, $(s^x)^T \nabla c(x) = 0$ so we also have $(s^x)^T \nabla_x L(x, p) = (s^x)^T \nabla_x f(x)$. So (3) are the KKT conditions of

(4)
$$\boxed{\begin{aligned} &\min_{s} \quad f(x_k) + s^T \nabla f(x_k) + \frac{1}{2} s^T \nabla_{xx}^2 L(x_k, p_k) s \\ &\text{s.t.} \quad s^T \nabla c_i(x_k) + c_i(x_k) = 0, \quad i \in \mathcal{E} \\ &\qquad \left(s^T \nabla c_i(x_k) + c_i(x_k) \geq 0, \quad i \in \mathcal{I} \right) \end{aligned}}$$

The Basic SQP Algorithm

Given : $x_0, p_0, (B_0)$ ← How? What?

set : $k \leftarrow 0$

while convergence not satisfied

Evaluate $f(x_k), \nabla f(x_k), c(x_k), \nabla c(x_k)$

Evaluate / Estimate $\nabla^2_{xx} L(x_k, p_k)$ ← q-Newton

solve (4) for s^*, λ^* ← QP solver

$x_{k+1} = x_k + s^*$

$p_{k+1} = -\lambda^*$

$k \leftarrow k+1$

} actually a line search with constraint violation penalty.

end while

Line Search SQP Algorithm

Given : $\eta \in (0, 1/2)$, $\tau \in (0,1)$, (x_0, λ_0) , $\rho \in (0,1)$

Evaluate : $f_0 = f(x_0)$, $\nabla f_0 = \nabla f(x_0)$, $c_0 = c(x_0)$, $\nabla c_0 = \nabla c(x_0)$

set : $k \leftarrow 0$, $B_0 = I_n$

while convergence not satisfied

$$\boxed{\begin{array}{l}\text{compute } s_k^x \text{ and } \lambda \text{ from } (4) \\ \text{set } s_k^p = \lambda - \lambda_k \ (p_k)\end{array}}$$

QP subproblem

$$\boxed{\begin{array}{l}\text{choose } \mu_k \geq \dfrac{\nabla f_k^T s_k^x + \frac{1}{2}(s_k^x)^T B_k s_k}{(1-\rho)\ \|c_k\|_1} \\ \\ \text{set } \alpha_k \leftarrow 1 \\ \text{while } \phi(x_k + \alpha_k s_k^x ; \mu_k) > \phi(x_k ; \mu_k) + \eta \alpha D(\phi(x_k ; \mu_k) s_k^x) \\ \qquad \alpha_k \leftarrow \tau_k \alpha_k \ , \quad \tau_k \in (0, \tau] \\ \text{end while} \\ x_{k+1} \leftarrow x_k + \alpha_k s_k^x \\ \lambda_{k+1} \leftarrow \lambda_k + \alpha_k s_k^p \\ (p_{k+1} \leftarrow p_k + \alpha_k s_k^p)\end{array}}$$

Line Search

Evaluate f_{k+1} , ∇f_{k+1} , c_{k+1} , ∇c_{k+1}

$$\boxed{\begin{array}{l} r_k \leftarrow \alpha_k s_k^x \\ y_k \leftarrow \nabla_x L(x_{k+1}, \lambda_{k+1}) - \nabla_x L(x_k, \lambda_k) \\ B_{k+1} \leftarrow \text{q-Newton update}\end{array}}$$

q-Newton update

end while

Some Details on the QP solve

$$\min_s \; f_k + s^T \nabla f_k + \frac{1}{2} s^T B_k s$$

$$\text{s.t.} \quad s^T \nabla c_i(x_k) + c_i(x_k) = 0 \;, \quad i \in \mathcal{E}$$

$$s^T \nabla c_i(x_k) + c_i(x_k) \geq 0 \;, \quad i \in \mathcal{I}$$

For solving $\min \; \frac{1}{2} s^T G s + s^T c$ ← not the same

$$\text{s.t.} \quad Ax \geq b$$

$$A_e x = b_e$$

Identifications:

$$G = B_k$$

$$c = \nabla_x f(x)$$

$$A = \nabla_x c_I(x)^T$$

$$b = -c_I(x)$$

$$A_e = \nabla_x c_E(x)^T$$

$$b_e = -c_E(x)$$

$$c_E(x) = [\cdots \; c_i(x) \; \cdots]^T \;, \quad i \in \mathcal{E}$$

$$\nabla_x c_E(x) = [\cdots \; \nabla c_i(x) \; \cdots] \;, \quad i \in \mathcal{E} \qquad (J(c_E(x))^T)$$

$$c_I(x) = [\cdots \; c_i(x) \; \cdots]^T \;, \quad i \in \mathcal{I}$$

$$\nabla_x c_I(x) = [\cdots \; \nabla c_i(x) \; \cdots] \;, \quad i \in \mathcal{I}$$

Value Design Project

	2.81	1.62	1.22	93.4	
(1)	2.667	1.747	1.411	[95]	93.6
(2)	2.40	1.73	1.48	[96]	93.8
(3)	3.88	2.14	1.18	[74 (16)]	38.1
(4)	3.25	1.50	1.50		
(5)	5.00	1.00	1.00		

The Subproblem may be infeasible

Rely on a relaxed feasible problem:

$$\begin{aligned} \min\ & f(x) \\ \text{s.t.}\ & c_i(x) = 0, \quad i \in \mathcal{E} \\ & c_i(x) \geq 0, \quad i \in \mathcal{I} \end{aligned} \quad \rightarrow \quad \begin{aligned} \min\ & f(x) + \mu \sum_{i \in \mathcal{E}} e_i + d_i + \mu \sum_{i \in \mathcal{I}} b_i \\ & c_i(x) = e_i - d_i, \quad i \in \mathcal{E} \\ & c_i(x) = -b_i, \quad i \in \mathcal{I} \\ & e, d, b \geq 0 \end{aligned}$$

Some Details on q-Newton Update — Damped BFGS —

$$r_k = x_{k+1} - x_k = \alpha_k s_k^x$$

$$y_k = \nabla_x L(x_{k+1}, \lambda_{k+1}) - \nabla_x L(x_k, \lambda_{k+1})$$

$$u_k = \theta y_k + (1-\theta) B_k s_k$$

$$\theta = \begin{cases} 1 & \text{if } r_k^T y_k \geq \frac{1}{5} r_k^T B_k r_k \\ 4/5\,(r_k^T B_k r_k)/(r_k^T B_k r_k - r_k^T y_k) & \text{otherwise} \end{cases}$$

$$B_{k+1} = B_k - (B_k r_k r_k^T B_k)/r_k^T B_k r_k + (u_k u_k^T)/r_k^T u_k$$